Research in Microbiology 174 (2023) 104039

Contents lists available at ScienceDirect

Research in Microbiology

journal homepage: www.elsevier.com/locate/resmic

Original Article

Identification of key amino acid residues in OqxB mediated efflux of fluoroquinolones using site-directed mutagenesis

Purnendu Bhowmik [a, b], Nagakumar Bharatham [a, b], Satoshi Murakami [c], Vasanthi Ramachandran [a, b], Santanu Datta [a, *]

[a] Bugworks Research India Pvt. Ltd., Centre for Cellular and Molecular Platforms, GKVK, Bellary Road, Bengaluru 560065, Karnataka, India
[b] The University of Trans-Disciplinary Health Sciences and Technology (TDU), 74/2, Post Attur via Yelahanka, Bengaluru, Karnataka 560064, India
[c] Department of Life Science and Technology, Tokyo Institute of Technology, Yokohama 226-8503, Japan

ARTICLE INFO

Article history:
Received 17 November 2022
Accepted 26 January 2023
Available online 3 February 2023

Keywords:
OqxB
Site-directed mutagenesis (SDM)
Klebsiella pneumoniae
Antimicrobial resistance
Efflux pumps

ABSTRACT

OqxB belongs to the RND (Resistance-Nodulation-Division) efflux pump family, recognized widely as a major contributor towards enhancing antimicrobial resistance. It is known to be predominantly present in all *Klebsiella* spp. and is attributed for its role in increasing resistance against an array of antibiotics like nitrofurantoin, quinolones, β-lactams and colistin. However, the presence of *oqxB* encoding this efflux pump is not limited only to *Klebsiella* spp., but is also found to occur via horizontal gene transfer in other bacterial genera like *Escherichia coli*, *Enterobacter cloacae* and *Salmonella* spp. Recently, we reported the crystal structure of OqxB and its structure-function relationship required for the efflux of fluoroquinolones. Extending these findings further, we characterized the structural architecture of this efflux pump along with identifying some critical amino acids at the substrate binding domain of OqxB. Based on our in silico modelling studies, both hydrophobic residues (F180, L280, L621, F626) and polar residues (R48, E50, E184, R157, R774) were found to be located at this site. The present work reports the importance of these key amino acid residues and the crucial ion–pair interactions at the substrate-binding pocket, thereby establishing their role in OqxB mediated efflux and the resultant resistance development against fluoroquinolones.

1. Introduction

The emerging threat of antimicrobial resistance (AMR) has become a matter of global concern. As recognised by the World Health Organization (WHO), AMR has become one of the most alarming health-care associated threats that humanity is facing today [1]. Recent studies have unravelled the enormity of the situation, as the current rate of progression of AMR is expected to rise to an annual death toll of ~10 million by 2050 [2]. In fact, the severity of the situation has only worsened in recent times owing to the unprecedented Covid-19 pandemic and the secondary infections-associated mortality [3]. Although there are a wide variety of microbes which are rapidly developing resistance and the related complications, most notorious among these belong to the ESKAPE group of pathogens (comprising of Gram-positive bacteria *Enterococcus faecium* and *Staphylococcus aureus;* and Gram-negative bacteria including *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, *Enterobacter* sp.) [4]. Among these organisms, *K. pneumoniae* has been found to be the most frequently encountered organism, associated with a wide variety of drug resistant nosocomial infections and the consequent deaths. *K. pneumoniae* is an opportunistic pathogen associated with hospital-borne infections like pneumonia, urinary tract infections (UTIs), bloodstream infections, wounds, surgical site infections etc., [5–7]. A recent study also revealed that a significant sub-group of Covid-19 patients, who suffered from secondary infections and exhibited a mortality rate of over 50%, were infected predominantly with multidrug-resistant *K. pneumoniae* [3].

The phenomenon of AMR is usually attributed to the enhanced resistance that the bacteria develop against one or more classes of antibiotics, which otherwise would have been susceptible and effective [8]. Bacteria could employ diverse strategies, like target

* Corresponding author.
E-mail addresses: purnendu@bugworksresearch.com (P. Bhowmik), nbharatham@bugworksresearch.com (N. Bharatham), murakami@bio.titech.ac.jp (S. Murakami), vasanthi@bugworksresearch.com (V. Ramachandran), santanu@bugworksresearch.com (S. Datta).

https://doi.org/10.1016/j.resmic.2023.104039
0923-2508/

gene mutations, drug inactivation, reduced entry or increased efflux, to overcome the antibiotic effect and thus become resistant [9]. Among all these mechanisms, the contribution of efflux pumps are of paramount importance, as these are among the most encountered causes behind the emergence of multidrug resistance [10–12]. Although bacteria possess a wide variety of efflux pumps [13], the members belonging to the RND (Resistance Nodulation Division) family are most widely prevalent [14]. Similar to other bacteria, the increased resistance exhibited by *K. pneumoniae* against common antibiotics like β-lactams, quinolones, nitrofurantoin, aminoglycosides and colistin, which were otherwise found effective, has been attributed to presence of efflux pumps [15,16]. Although a diverse array of RND efflux pumps have been reported in *K. pneumoniae*, the OqxAB and AcrAB are the predominant ones, identified for their roles in developing resistance to multiple classes of antibiotics [15].

OqxB belongs to the RND efflux pump family and is predominantly considered for its role in conferring resistance across a wide variety of antibiotics, like quinolones, nitrofurantoin, chloramphenicol, quinoxalines, tigecycline, disinfectants and detergents [15–22]. Although *oqxB*, the gene encoding this efflux pump is present on the bacterial chromosome, the presence of the gene additionally on transmissible plasmids and mobile insertion elements like IS26 makes it a potential reservoir of antimicrobial resistance for movement across to other bacterial genera [22]. Studies have shown the existence of this plasmid in clinical isolates of *K. pneumoniae*, *E. coli* and *E. cloacae*, contributing to multi-drug resistance against commonly used antibiotics, further indicating the role of this efflux pump in driving AMR across isolates [23]. Recently, we reported the crystal structure of OqxB at 1.85 Å resolution, along with the insights of its unique substrate-binding site and its contribution for the efflux of fluoroquinolones [24]. Although the overall structural assembly of OqxB is similar to the well-studied and characterized RND efflux pump components from other bacterial systems, like AcrB in *E. coli* and MexB in *P. aeruginosa*, the distinctive architecture of its substrate-binding site makes it unique and versatile. The substrate-binding domains of well characterized RND pumps of the bacterial systems like AcrB in *E. coli* and MexB in *P. aeruginosa*, are primarily composed of an aromatic phenylalanine cage, along with other non-polar hydrophobic residues (like tyrosine, alanine, leucine, isoleucine and valine) or uncharged polar residues (like serine and asparagine). In contrast, the substrate-binding domain of OqxB is made up of distinct residues. In addition to some of the conserved residues of the phenylalanine cage, it is made up of hydrophobic (phenylalanine, alanine, valine, leucine), polar uncharged (glutamine) and charged (aspartate, glutamate, arginine and lysine) amino acids. The detailed structural and functional analysis provided here offers some insights about the features of the substrate-binding pocket that leads to the likely consequence of efflux of fluoroquinolones. Our earlier studies based on docking and molecular dynamics simulations indicated that the interactions of different fluoroquinolone molecules with both the polar as well as non-polar residues at the substrate-binding pocket are crucial for their successful binding and subsequent extrusion [24].

In the present study, we report the contributions of key amino acid residues of the substrate binding domain of OqxB, as predicted by docking studies and MD simulations from our earlier work [24] in mediating efflux of fluoroquinolones. Moreover, with the studies conducted, we could also establish that interactions of the charged substitutions on substrate with the polar/charged groups of phenylalanine cage were equally crucial along with that of the hydrophobic interactions with the non-polar hydrophobic residues, for the imminent efflux of fluoroquinolones. In order to understand the role of specific residues of OqxB in mediating latter, MICs were determined against different fluoroquinolones following over-expression of both wild type and mutated *K. pneumoniae* OqxB generated via site directed mutagenesis (SDM) and transformation of *E. coli*. We propose that the results from this study could lead to a better understanding of the mode of efflux mechanisms employed by OqxB and consequently yield valuable insights on how to counter the associated challenge of AMR.

2. Materials and methods

2.1. Preparation of ligands and proteins for docking simulations

The preparation of ligands and proteins for docking studies was done, as mentioned earlier [24]. Briefly, the Ligprep [25] module in the Schrodinger suite was used to prepare the 2D structures of olaquindox (OLA), ciprofloxacin (CIPRO), levofloxacin (LEVO), and moxifloxacin (MOXI). Ligprep accepts 2D molecules and converts them to 3D. The Ligprep Epik sub-module was used to generate tautomers and potential ionisation states for each molecule. Each ligand was manually examined to ensure that it was in the proper tautomer and ionisation state at physiologically relevant pH (7.4). The MacroModel-MTLM (mixed torsional/low-mode) method was used to generate multiple conformations from Ligprep output structures. MacroModel-MTLM combines a Monte Carlo torsional space exploration method (which efficiently locates widely separated minima on a potential energy surface) with a low-mode conformational search method that searches along energetically “soft” degrees of freedom. The OPLS 2005 force field was used, and the TNCG method was used to minimise energy for 500 steps. The acceptable structure energy window was set at 21 kJ/mol (default value). Conformations of the same molecule that were within 0.5 RMSD of each other were culled. For Monte Carlo sampling, a maximum of 500 steps were allowed, and for low mode searching, a maximum of 50 steps were allowed. A maximum of 20 conformers were allowed per molecule.

2.2. Molecular docking

Studies on molecular docking were performed as discussed previously [24]. The generation of unbiased binding modes for olaquindox and other fluoroquinolones derivatives (ciprofloxacin, moxifloxacin and levofloxacin) were done using the Glide v8.5 [26] molecular docking module (Schrodinger, 2019) in SP-mode. As all the chains are highly similar, only one out of three chains were considered for docking. Using the Protein Preparation Wizard in the Maestro suite from Schrodinger, the protein with the ligand molecule was processed initially to add the correct protonation state and bond orders to hetero atoms of protein residues, water molecules and the bound ligands. This processed complex was used to generate the pre-computed docking grid. The docking protocol started with the ligand's systematic conformational expansion, followed by placement in the receptor site. Minimization of the ligand in the receptor's field was then carried out using the OPLS-AA force field with the default distance-dependent dielectric. The lowest energy poses were then subjected to a Monte Carlo procedure that sampled nearby torsional minima. Poses were ranked using GlideScore, a modified version of the ChemScore function that includes terms for steric clashes and buried polar groups. The default van der Waals's scaling was used (1.0 for the receptor and 0.8 for the ligand). The resulted dock poses were visualized manually in PyMOL software.

2.3. Bacterial strains

All the bacterial strains used in the present study are listed in Supplementary Table S1. The strains were grown and cultured on either 1.5% Luria Bertani Agar Miller (Hi–Media) plates or in Mueller Hinton Broth No.2 Control Cations (Hi–Media). Selection pressure with antibiotics like ampicillin (Ampicillin sodium salt; Sigma–Aldrich, Catalog no.# A9518), amp 100 μg/mL or kanamycin (Kanamycin sulfate; HiMedia, Catalog no.# MB105), kan 30 μg/mL, were used, wherever required. Double knockout *ΔacrA-ΔacrB* strain in *E. coli* BW25113 was generated by homologous recombination, as discussed elsewhere [27]. *E. coli* Top10 F' strain (ThermoFisher Scientific) was used for DNA cloning and plasmid amplification. All the cultures were grown at 37 °C under aerobic conditions.

2.4. Plasmids and constructs

The plasmids used in the current study are listed in Supplementary Table S2. Each of the constructs were designed and made, as discussed previously [24] for the heterologous expression of the efflux pump components. The polycistronic *acrAB* and *oqxAB* were amplified using genomic DNA templates from *E. coli* BW25113 and *K. pneumoniae* (JCM1662) genomic DNA, respectively, using high fidelity Phusion DNA Polymerase (ThermoFisher Scientific). The amplicons for the complementation studies for *acrA-acrB* and *oqxA-oqxB*, along with a constitutive promoter sequence, were fused with a vector backbone containing the *pBR322*-origin and Ampicillin resistance gene (from the vector pTrc99a, Genscript) using the *NdeI* and *HindIII* restriction sites. The strains used for the complementation assays were generated by transforming these plasmids as described in Supplementary Table S1.

2.5. Site-directed mutagenesis (SDM) for OqxB

The contribution of the polar residues at the substrate binding site of OqxB in the efflux of fluoroquinolones, was studied by generating and using mutants, in which the polar residues were replaced with non-polar alanine residue. The strategy followed for generating these mutants is same, as discussed previously [24]. Briefly, the mutation was introduced by site-directed mutagenesis by a method based on the process of rolling circle PCR amplification of the template, as developed by Stratagene (Catalog #200518) [28]. Each of the desired mutations were introduced in *oqxB* gene, by amplifying the plasmid carrying genes *oqxA-oqxB*, using mutagenic primers, as described in Supplementary Table S3. The synthesis of the mutant strand was carried out in a Thermal Cycler (Biorad) with the cycling program of 1 hold of 95 °C for 5min, 15 cycles of 95 °C for 30 s, 55 °C for 1min, 68 °C for 6 min and final extension of 68 °C for 10min. Subsequently, the amplified product was digested with *DpnI* (ThermoFisher Scientific) at 37 °C for 1 h to eliminate methylated plasmid substrate. The mixture was then transformed into *E. coli* Top10 F' competent cells and selected on ampicillin (100 μg/mL) containing plates. Plasmids from individual colonies were screened for the presence of desired mutations, by sequencing. Following confirmation for the presence of the mutation, the plasmids were transformed in *E. coli* BW25113 strains and subsequently used for complementation assays.

2.6. Determination of minimal inhibitory concentration (MIC)

MIC was determined, as per the guidelines of CLSI, as discussed elsewhere [29,30]. All the strains were streaked on LB-Agar plates with either ampicillin (amp 100 μg/mL) or kanamycin (kan 30 μg/mL), for the respective strains carrying the selection marker, from glycerol stocks maintained at −80 °C and allowed to grow overnight at 37 °C. A single colony of each strain was picked from the fresh plates and grown in MHB No.2 Control Cations (Hi–Media) with the selected antibiotic, ampicillin (100 μg/mL) or kanamycin (30 μg/mL). As both AcrAB and OqxAB expression are under a constitutive promoter, no additional induction was required for triggering their expression. Olaquindox (Sigma–Aldrich; Lot#BCBW1743), ciprofloxacin (Ciprofloxacin HCL; Sigma–Aldrich; Lot#LRAB3671), norfloxacin (Sigma–Aldrich; Lot#SLCD0585), levofloxacin (Sigma, Lot#BCBF7004V) and moxifloxacin (Moxifloxacin HCl; Sigma–Aldrich, Lot#LRAA7396) were procured and used in this study. All the antibiotic stock solutions and their two-fold dilutions were prepared in DMSO. 3 μl of each compound dilutions were added to 150 μL of bacterial culture (with ($3–7 \times 10^5$ CFU/mL), in a flat bottom 96-well microtiter plates, for obtaining final antibiotic concentrations ranging from 160 μg/mL to as low as 0.03 μg/mL. The plates were packed in gas permeable polythene bags and incubated overnight (16–18 h), at 37 °C. The next day, absorbance was measured for each of the wells at 600 nm using a spectrophotometer and MIC was determined as the concentration at which no bacterial growth was observed. For wild type strains, a cut-off of 90% growth inhibition was used for MIC determination. However, in the recombinant strains, as the growth is compromised partially due to the presence of a plasmid and overexpression of target genes, a cut-off of 80% growth inhibition with respect to their control levels was used for determining the MICs. All experiments were performed in triplicates and the data reported in the present work is representative of the same.

3. Results and discussion

Prior to the crucial event of the drug inhibiting its intracellularly located target enzyme that leads to the antibacterial effect, two key features, permeability and efflux govern the availability of the drug within the cell. Role of multiple factors including passive diffusion, membrane translocators, specific transporters, porins and efflux pumps define the actual concentration for this interaction [31]. Thus, apart from the intended target, the drug comes in contact with other proteins which could dictate the extent of antimicrobial effect. The final outcome is also governed by the affinity to the specific target and the vulnerability within the milieu. Earlier work by Vergalli et al., have captured comprehensively the similarities and differences among these key parameters and have categorized multiple fluoroquinolones keeping the drug features (rotatable bonds, shape) into consideration versus their affinity, extent of influx and efflux [32]. The focus of this report is on the latter, specifically with the aim to gain an understanding of emergence of OqxB mediated efflux in *K. pneumoniae* and its likely role in conferring resistance to standard of care. Therefore, we have selected only four representative fluoroquinolone molecules (ciprofloxacin, levofloxacin, norfloxacin and moxifloxacin) for this study.

Previously, we reported the crystal structure and the detailed structure-function relationship of the RND efflux pump OqxB, from *K. pneumoniae* [24]. Our earlier work provided an insight into the unique structural features of the substrate binding domain of the phenylalanine cage of OqxB, and their probable roles in the efflux of a wide spectrum of antibiotics, including the fluoroquinolones. Based on findings from our in silico studies using molecular docking and MD simulations, and the comparison of their structural features with that of the well-studied efflux pumps like AcrB and MexB, putative amino acid residues likely to have a critical impact around the n-dodecyl-β-D-maltoside (DDM) binding pocket of OqxB were identified. Extending this understanding further, we identified and confirmed the roles of the predicted residues in the binding and subsequent efflux of quinoxalines, for e.g., olaquindox and various fluoroquinolones such as ciprofloxacin, norfloxacin, levofloxacin, and moxifloxacin.

As evident from several structural and mutant generation studies reported earlier, F178 in AcrB and MexB, is one of the key and central aromatic residue which interacts with most of the antibiotic/substrate molecules [33,34]. Moreover, our earlier investigations with nitro containing compounds also revealed that I277 is a key residue in the hydrophobic substrate binding pocket [29]. Correspondingly, in case of OqxB, however, an identical residue F180 is present instead of F178 of AcrB and MexB, along with a similar hydrophobic residue L280 which is present instead of I277. The predicted binding modes of olaquindox and fluoroquinolones (Fig. 1) showed that these two residues are likely to be involved in the substrate binding. The g-loop (gate loop), which also plays a key role in permeation into substrate binding pocket or phenylalanine cage, also demonstrated differences with AcrB and MexB (PDB ID: 4DX5 and 3W9I, respectively) in studies reported earlier [24]. In comparison to AcrB and MexB, the g-loop is extended by two extra residues, and additionally, the key F615 and F617 residues, are replaced by L621 and A623 in OqxB. Surprisingly, our DDM bound OqxB structure (PDB ID: 7CZ9) indicated that the orientation of F626 side chain is such that it occupies the space of F617 of AcrB and MexB, thereby plausibly playing a similar role in substrate binding/positioning in phenylalanine cage.

Though all four fluoroquinolone molecules contain key common features, such as the negatively charged carboxyl group on one side and the positively charged amine group on the other, few differences were observed based on their binding, especially along the g-loop interacting portions of the molecules. Ethyl and cyclopropyl substitutions were present on norfloxacin and ciprofloxacin molecules, respectively, whereas an extra methoxy group was present on the bicyclic ring of moxifloxacin. In contrast, levofloxacin consisted of a fused ring in place of these substitutions (Supplementary

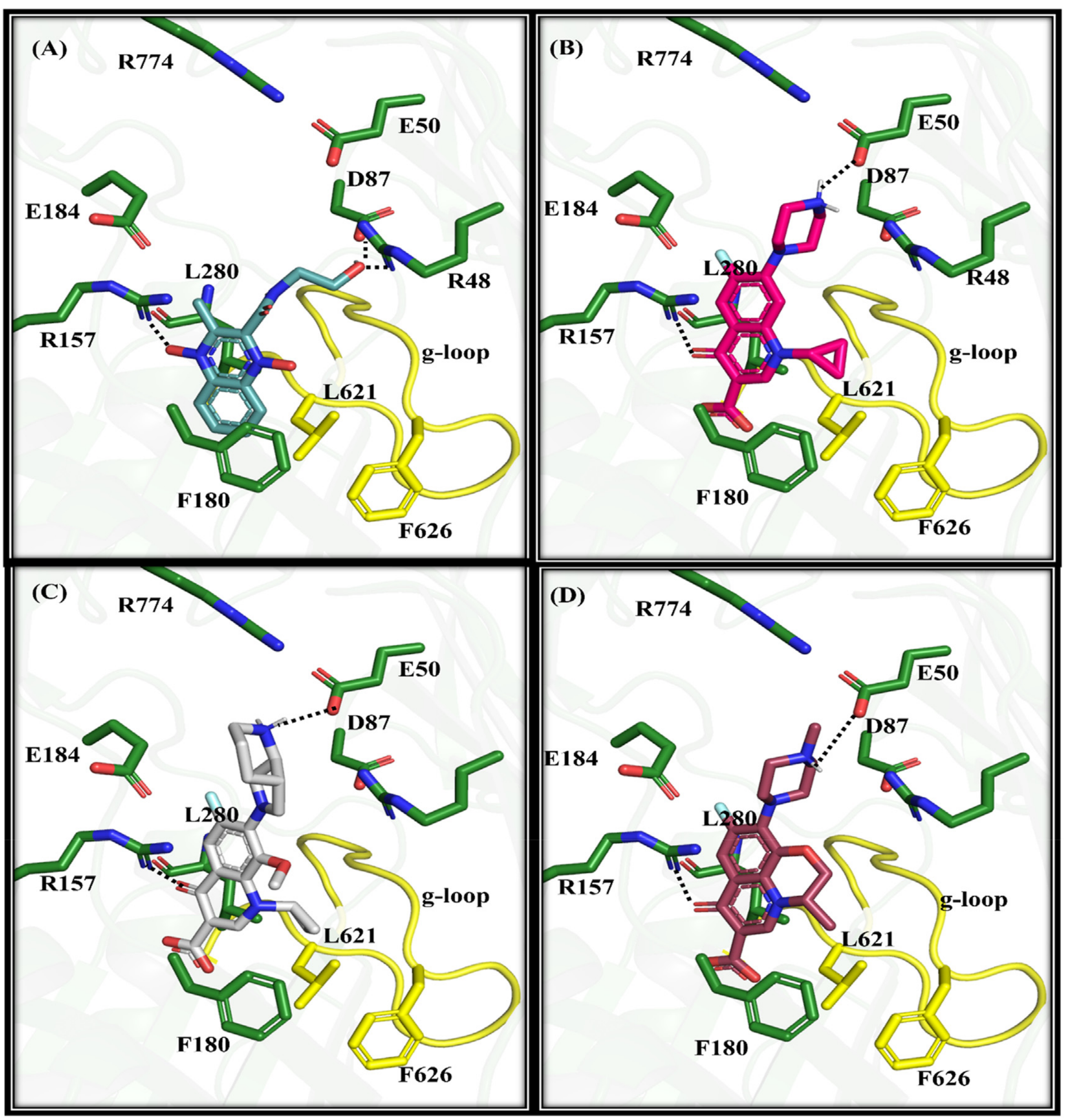

Fig. 1. Molecular Docking for the predicted binding modes of olaquindox (A) ciprofloxacin (B), moxifloxacin (C), and levofloxacin (D) at the substrate binding pocket or phenylalanine cage of OqxB. Olaquindox depicted as dark cyan whereas all the three fluoroquinolones: ciprofloxacin (magenta), moxifloxacin (grey) and levofloxacin (brown) sticks. Important interacting residues as well as ion-pairs shown as dark green sticks and hydrogen bond interactions between protein and ligand shown as broken lines. The g-loop (gate loop) which is key for substrate entry into phenylalanine cage highlighted in yellow colour and the two key hydrophobic residues shown as sticks.

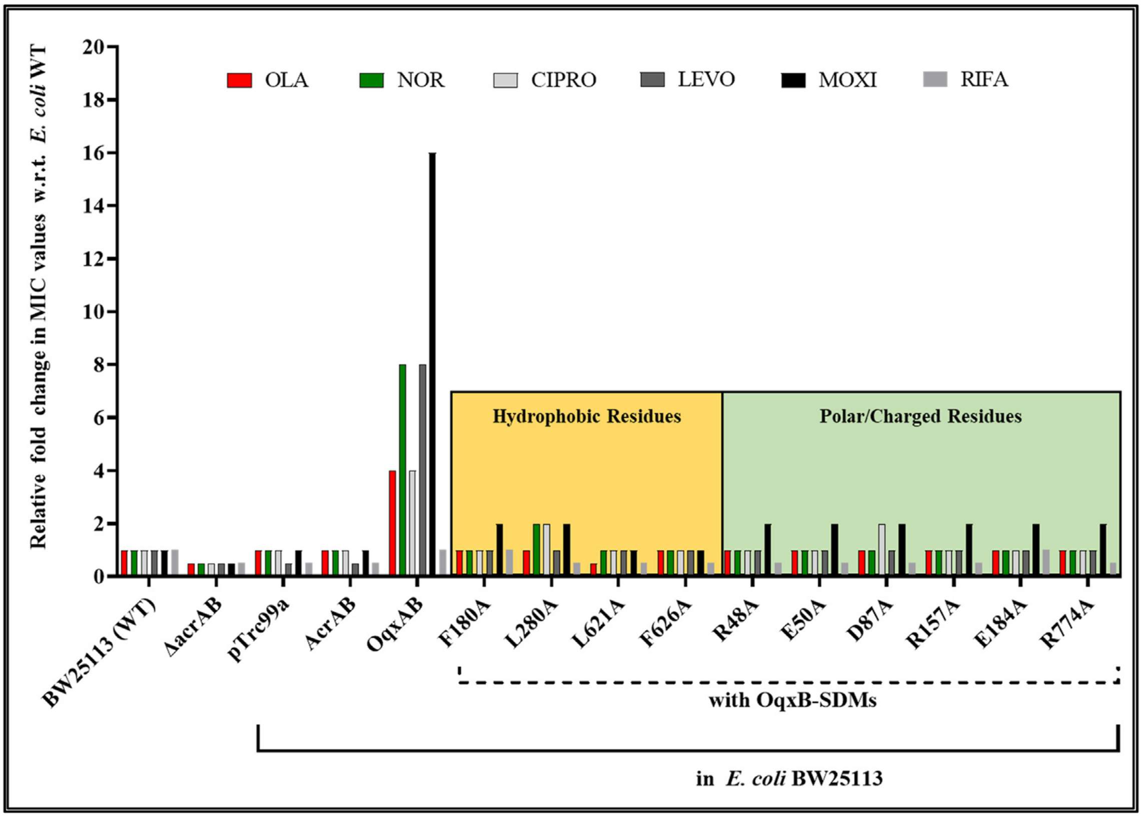

Fig. 2. Relative Fold-change in MIC values. Relative fold change for the MIC values of olaquindox and different fluoroquinolones in *E. coli* BW25113 (WT) complemented with either *K. pneumoniae* OqxA-OqxB or OqxA-OqxB (with SDMs) overexpression systems; Rifampicin was used as a negative control. pTrc99a, +AcrAB, +OqxAB, +OqxA + OqxB(SDMs): complementation with plasmid carrying the respective genes or the gene with indicated mutations. **OLA**: Olaquindox; **NOR**: Norfloxacin; **CIPRO**: Ciprofloxacin; **LEVO**: Levofloxacin; **MOXI**: Moxifloxacin; **RIFA**: Rifampicin.

Table 1
MIC values of OqxA-OqxB overexpressing complementation systems in *E. coli* BW25113. The minimal inhibitory concentrations of *E. coli* WT systems overexpressed with either OqxAB or OqxA-OqxB (with SDM), against quinoxaline and other different fluoroquinolones; rifampicin was used as a negative control.

Strains	MIC values (in μg/ml)					
	OLA	NOR	CIPRO	LEVO	MOXI	RIFA
E. coli K-12 BW25113 (WT)	40	0.05	0.0125	0.05	0.025	10
ΔacrAB	20	0.025	0.00625	0.025	0.0125	5
WT + pTrc99a	40	0.05	0.0125	0.025	0.025	5
WT + AcrAB	40	0.05	0.0125	0.025	0.025	5
WT + OqxAB	160	0.4	0.05	0.4	0.4	10
WT + OqxA + OqxB(F180A)*	40	0.05	0.0125	0.05	0.05	10
WT + OqxA + OqxB(L280A)*	40	0.1	0.025	0.05	0.05	5
WT + OqxA + OqxB(L621A)*	20	0.05	0.0125	0.05	0.025	5
WT + OqxA + OqxB(F626A)*	40	0.05	0.0125	0.05	0.025	5
WT + OqxA + OqxB(R48A)#	40	0.05	0.0125	0.05	0.05	5
WT + OqxA + OqxB(E50A)#	40	0.05	0.0125	0.05	0.05	5
WT + OqxA + OqxB(D87A)#	40	0.05	0.025	0.05	0.05	5
WT + OqxA + OqxB(R157A)#	40	0.05	0.0125	0.05	0.05	5
WT + OqxA + OqxB(E184A)#	40	0.05	0.0125	0.05	0.05	10
WT + OqxA + OqxB(R774A)#	40	0.05	0.0125	0.05	0.05	5

E. coli K-12 BW25113 (WT: wildtype) or its *ΔacrAB* strain used for MIC.
pTrc99a, +AcrAB, +OqxAB, +OqxA + OqxB(SDMs): complementation with plasmid carrying the respective genes or the gene with indicated mutations.
OLA: Olaquindox; **NOR**: Norfloxacin; **CIPRO**: Ciprofloxacin; **LEVO**: Levofloxacin; **MOXI**: Moxifloxacin; **RIFA**: Rifampicin.
*SDM in hydrophobic residues; # SDM in polar/hydrophilic residues.

Fig. S1). Preliminary results from our studies indicated that the altered side chains facilitate better hydrophobic interactions with g-loop residues like L621 (Fig. 1). Unlike *E. coli* AcrB, where phenylalanine cage residues like F136, F178, F610, F615 and F628 are major contributors to the binding and recognition of fluoroquinolones, OqxB uses a combination of hydrophobic and charged residues for recognition. The zwitterionic nature of fluoroquinolone compounds complements well in the OqxB pocket, which might be crucial for binding to the OqxB pocket and subsequent extrusion by the efflux pump. However, further in-depth studies, along with structural and functional explorations, are needed to be done for the detailed assessment of the entry, accumulation, and efflux of fluoroquinolones by OqxB.

In order to validate the roles of crucial hydrophobic residues of OqxB in quinolone binding, site-directed mutagenesis of the four residues (F180, L280, L621 and F626), and MIC based complementation assays were conducted. For these studies, the heterologous system comprising of *K. pneumoniae* wildtype or mutated OqxB, was constitutively overexpressed in *E. coli* BW25113. Since most of the studies using clinical isolates of *Klebsiella* spp., *E. coli* and *Salmonella* spp. that exhibit fluoroquinolone resistance are usually found to be associated with the presence or transmission of plasmid-mediated quinolone resistance (PMQR) of OqxB via horizontal gene transfer [24,35,36], *E. coli* wildtype was chosen as the host strain for our studies, instead of the efflux pumps deleted double knockout strains (*ΔacrAB*), so as to mimic the possible scenario frequently encountered among the clinical isolates in the human host environment. For the complementation studies, host cells transformed with pTrc99a served as the control system. Surprisingly, *E. coli* BW25113 *ΔacrAB* transformed with the native *E. coli* AcrA-AcrB showed insignificant impact with regards to the efflux of fluoroquinolones. In contrast, complementation studies with wildtype and mutated *Klebsiella pnuemoniae* OqxB, in *E. coli*

BW25113 system exhibited an altered resistance pattern with specificity towards fluoroquinolones. The overexpression of wild-type OqxA-OqxB in *E. coli* significantly enhanced the MIC values, resulting in an enhanced shift of 8–16 fold (in comparison to wild type *E. coli*). However, substitution of the hydrophobic residues with alanine using site directed mutagenesis, resulted in complete reversion of the fluoroquinolone extrusion ability of OqxB efflux pump, as evidenced from the restored MIC values, similar to that seen with wild type *E. coli* BW25113 (Fig. 2, Table 1). These observed outcomes are also in agreement with the previously published work on OqxB, wherein involvement of residues F180 and F626 in binding with compounds like olaquindox and ciprofloxacin, was reported in *E. coli* BL21 [37]. In contrast, the overexpression of the native AcrA-AcrB in *E. coli* did not show any considerable shift in the MIC values for antibiotics like olaquindox, ciprofloxacin, norfloxacin, moxifloxacin, and levofloxacin, further confirming the specificity of OqxB in extrusion of fluoroquinolones (Table 1).

In addition to the conventional hydrophobic/aromatic residues in phenylalanine cage, OqxB pocket also possesses few hydrophilic residues, including the three ion pairs of complementary charged residues (E50:R774, D87:R48, and R157:E184). These ion-pairs within the substrate binding pocket are unique to OqxB and has not been found in any other efflux pumps so far. The guanidino side chain of the residue R157, orients towards the substrate binding pocket and has been found to interact with carbonyl group of quinolones, as observed in docking simulations. Furthermore, MD simulations also revealed the additional and stable interactions of R157 with carbonyl and terminal carboxyl group of quinolone derivatives. Another small hydrophilic residue S155, which is present in both AcrB and MexB efflux pumps, is replaced with the positively charged arginine (R157) in OqxB [24]. Although the substrate binding pocket of the well characterized RND efflux pumps like AcrB and MexB, are majorly composed of hydrophobic/aromatic residues, hydrophilic residues like serine and threonine are also found to be present at upper portion of the cage or at the opening of the exit tunnel. In contrast, these hydrophilic residues are replaced, with two sets of complimentary charged residues (D87: R48 and E50:R774), in the corresponding pocket of OqxB. The negatively charged side chains of E50 and D87 residues are predicted to be crucial in interacting with positively charged substitutions of quinolones and terminal hydroxyl group of olaquindox, respectively. Additionally, the guanidino side chain of R48 is also found to be interacting with the oxygen of hydroxyl group (Fig. 1). Although our earlier studies had confirmed the role of R157 in OqxB mediated efflux of fluoroquinolones [24], the contributions of the other predicted charged residues needed further validation. Therefore, to establish the validity of these predictions and assign the role of these unconventional ion pairs in efflux activity, we selected five additional charged residues (R48, E50, D87, E184 and R774) for site-directed mutagenesis (SDM) studies and complementation assays. As discussed earlier, the substitution of these polar residues with alanine, resulted in complete inactivation of the OqxB mediated efflux of fluoroquinolones, in *E. coli* BW25113 background, transformed with the heterologous *K. pneumoniae* OqxA-OqxB. The outcome of these studies are similar to that of a R157 mutation effect, as reported earlier [24]. The MIC values for olaquindox, moxifloxacin, levofloxacin, norfloxacin and ciprofloxacin showed a significant reduction of 4-8-fold, in comparison to that of wild type OqxA-OqxB overexpressing system, thus further confirming the role of these hydrophilic residues in the efflux of fluoroquinolones (Fig. 2, Table 1).

Overall, the reported residues in the substrate binding pocket of OqxB are crucial for the effective positioning, binding, and subsequent efflux of fluoroquinolones. The aromatic ring of the quinolone makes pi-cation as well as charged–charged or hydrogen bond interactions with the guanidino side chain of R157, whereas the residue E50 is associated with the H-bond interactions with the piperazine ring derivatives of the fluoroquinolones, thereby causing stabilization of its binding. The charged residue pairs (E50:R774 and R157:E184) are further involved in intra-molecular interactions and are crucial for stabilizing the binding of fluoroquinolones with the OqxB binding pocket. The hydrophobic interactions between the side chain of fluoroquinolones and the pocket comprising of the residues like F180, L280, L621and F626, further facilitates effective positioning and their binding to the substrate binding pocket of OqxB.

4. Conclusion

In summary, we report here the detailed characterization of the structural architecture of OqxB efflux pump and identification of unique amino acid residues within its substrate binding pocket. In the current study we have also identified the crucial interactions of olaquindox and various fluoroquinolones like ciprofloxacin, norfloxacin, levofloxacin and moxifloxacin at the substrate binding pocket. Further we established the importance of the positively charged residues (R48, R157 and R774) and the negatively charged residues (E50, D87 and E184). Contrary to the other RND efflux pumps, we report here that the effective binding of the substrate involves the combined contributions of the hydrophobic pocket with residues like F180A, L280, L621and F626, as well as the charged residues of OqxB. Their critical roles in the binding and efflux of fluoroquinolones based on the predictions by docking studies and MD simulations were ascertained and validated using site directed mutagenesis and MIC based complementation assays. This work implicated that the unique structural components of the OqxB efflux pumps could be the influencing factor, that comprises features such as combination of aromatic ring systems with charge, along with zwitterionic/hydrophilic substitutions which in turn dictates its ability to extrude certain classes of antibiotics. These are the key recognition features shared by such antibiotics as demonstrated in these studies. Observations from such investigations hold the promise of a build-up of knowledge base of the mechanisms employed by pumps like OqxB, which in turn could aid in design of agents countering efflux associated AMR.

Author's contribution

P.B. designed and performed the SDM, complementation and MIC experiments. N.B. designed and performed all the computational docking studies. V.R. and S.D. contributed to the overall designing of the studies. S.M. contributed to the conceptualization and analysis of the structural studies. N.B., P.B., V.R and S.D. wrote the manuscript. All authors revised and contributed to the manuscript.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors thank the Centre for Cellular and Molecular Platforms (C-CAMP), Bangalore, for Bio-Incubation facility support and Bugworks Research Inc. for supporting this work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.resmic.2023.104039.

References

[1] Murray CJ, Ikuta KS, Sharara F, Swetschinski L, Robles Aguilar G, Gray A, et al. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet 2022;399:629–55.

[2] de Kraker MEA, Stewardson AJ, Harbarth S. Will 10 million people die a year due to antimicrobial resistance by 2050? PLoS Med 2016;13:1002184.

[3] Vijay S, Bansal N, Rao BK, Veeraraghavan B, Rodrigues C, Wattal C, et al. Secondary infections in hospitalized COVID-19 patients: Indian experience. Infect Drug Resist 2021;14:1893–903.

[4] Pendleton JN, Gorman SP, Gilmore BF. Clinical relevance of the ESKAPE pathogens. Expert Rev Anti Infect Ther 2013;11:297–308.

[5] Zowawi HM, Forde BM, Alfaresi M, Alzarouni A, Farahat Y, Chong TM, et al. Stepwise evolution of pandrug-resistance in *Klebsiella pneumoniae*. Sci Rep 2015;5:1–8.

[6] Ferreira RL, Da Silva BCM, Rezende GS, Nakamura-Silva R, Pitondo-Silva A, Campanini EB, et al. High prevalence of multidrug-resistant *Klebsiella pneumoniae* harboring several virulence and β-lactamase encoding genes in a brazilian intensive care unit. Front Microbiol 2019;10:3198.

[7] Bassetti M, Righi E, Carnelutti A, Graziano E, Russo A. Multidrug-resistant *Klebsiella pneumoniae*: challenges for treatment, prevention and infection control. Expert Rev Anti Infect Ther 2018;16:749–61.

[8] Dhingra S, Rahman NAA, Peile E, Rahman M, Sartelli M, Hassali MA, et al. Microbial resistance movements: an overview of global public health threats posed by antimicrobial resistance, and how best to counter. Front Public Heal 2020;8:531.

[9] Abushaheen MA, Muzaheed, Fatani AJ, Alosaimi M, Mansy W, George M, et al. Antimicrobial resistance, mechanisms and its clinical significance. Disease-a-Month 2020;66:100971.

[10] Sun J, Deng Z, Yan A. Bacterial multidrug efflux pumps: mechanisms, physiology and pharmacological exploitations. Biochem Biophys Res Commun 2014;453:254–67.

[11] Hernando-Amado S, Blanco P, Alcalde-Rico M, Corona F, Reales-Calderón JA, Sánchez MB, et al. Multidrug efflux pumps as main players in intrinsic and acquired resistance to antimicrobials. Drug Resist Updat 2016;28:13–27.

[12] Blanco P, Hernando-Amado S, Reales-Calderon JA, Corona F, Lira F, Alcalde-Rico M, et al. Bacterial multidrug efflux pumps: much more than antibiotic resistance determinants. Microorganisms 2016;4.

[13] Piddock LJV. Multidrug-resistance efflux pumps - not just for resistance. Nat Rev Microbiol 2006;4:629–36.

[14] Alvarez-Ortega C, Olivares J, Martínez JL. RND multidrug efflux pumps: what are they good for? Front Microbiol 2013;4.

[15] Rodríguez-Martínez JM, de Alba PD, Briales A, Machuca J, Lossa M, Fernández-Cuenca F, et al. Contribution of OqxAB efflux pumps to quinolone resistance in extended-spectrum-β-lactamase-producing *Klebsiella pneumoniae*. J Antimicrob Chemother 2013;68:68–73.

[16] Ni RT, Onishi M, Mizusawa M, Kitagawa R, Kishino T, Matsubara F, et al. The role of RND-type efflux pumps in multidrug-resistant mutants of *Klebsiella pneumoniae*. Sci Rep 2020 2020;101:1–10. 10.

[17] Ho PL, Ng KY, Lo WU, Law PY, Lai ELY, Wang Y, et al. Plasmid-mediated OqxAB is an important mechanism for nitrofurantoin resistance in *Escherichia coli*. Antimicrob Agents Chemother 2016;60:537–43.

[18] Yuan J, Xu X, Guo Q, Zhao X, Ye X, Guo Y, et al. Prevalence of the oqxAB gene complex in *Klebsiella pneumoniae* and *Escherichia coli* clinical isolates. J Antimicrob Chemother 2012;67:1655–9.

[19] Hong BK, Wang M, Chi HP, Kim EC, Jacoby GA, Hooper DC. oqxAB encoding a multidrug efflux pump in human clinical isolates of Enterobacteriaceae. Antimicrob Agents Chemother 2009;53:3582–4.

[20] Xu Q, Jiang J, Zhu Z, Xu T, Sheng ZK, Ye M, et al. Efflux pumps AcrAB and OqxAB contribute to nitrofurantoin resistance in an uropathogenic *Klebsiella pneumoniae* isolate. Int J Antimicrob Agents 2019;54:223–7.

[21] Li J, Zhang H, Ning J, Sajid A, Cheng G, Yuan Z, et al. The nature and epidemiology of OqxAB, a multidrug efflux pump. Antimicrob Resist Infect Control 2019;8:1–13.

[22] Wyres KL, Holt KE. *Klebsiella pneumoniae* as a key trafficker of drug resistance genes from environmental to clinically important bacteria. Curr Opin Microbiol 2018;45:131–9.

[23] xin Zheng J, wei Lin Z, Sun X, hong Lin W, Chen Z, Wu Y, et al. Overexpression of OqxAB and MacAB efflux pumps contributes to eravacycline resistance and heteroresistance in clinical isolates of *Klebsiella pneumoniae*. Emerg Microbes Infect 2018;7.

[24] Bharatham N, Bhowmik P, Aoki M, Okada U, Sharma S, Yamashita E, et al. Structure and function relationship of OqxB efflux pump from *Klebsiella pneumoniae*. Nat Commun 2021;5400:1–12.

[25] Chen IJ, Foloppe N. Drug-like bioactive structures and conformational coverage with the LigPrep/ConfGen suite: comparison to programs MOE and catalyst. J Chem Inf Model 2010;50:822–39.

[26] Friesner RA, Banks JL, Murphy RB, Halgren TA, Klicic JJ, Mainz DT, et al. Glide: a new approach for rapid, accurate docking and scoring. 1. Method and assessment of docking accuracy. J Med Chem 2004;47:1739–49.

[27] Datsenko KA, Wanner BL. One-step inactivation of chromosomal genes in *Escherichia coli* K-12 using PCR products. *Proc Natl Acad Sci* U S A 2000;97:6640–5.

[28] Braman J, Papworth C, Greener A. Site-directed mutagenesis using double-stranded plasmid DNA templates. Methods Mol Biol 1996;57:31–44.

[29] Hameed P S, Bharatham N, Katagihallimath N, Sharma S, Nandishaiah R, Shanbhag AP, et al. Nitrothiophene carboxamides, a novel narrow spectrum antibacterial series: mechanism of action and Efficacy. Sci Rep 2018;81:1–18. 8.

[30] Ma W. Methods for dilution antimicrobial susceptibility tests for bacteria that grow aerobically : approved standard. CLSI 2006;26:M7–7.

[31] Rybenkov VV, Zgurskaya HI, Ganguly C, Leus IV, Zhang Z, Moniruzzaman M. The whole is bigger than the sum of its parts: drug transport in the context of two membranes with active efflux. Chem Rev 2021;121:5597.

[32] Vergalli J, Atzori A, Pajovic J, Dumont E, Malloci G, Masi M, et al. The challenge of intracellular antibiotic accumulation, a function of fluoroquinolone influx versus bacterial efflux. Commun Biol 2020;31:1–12. 3.

[33] Sjuts H, Vargiu AV, Kwasny SM, Nguyen ST, Kim HS, Ding X, et al. Molecular basis for inhibition of AcrB multidrug efflux pump by novel and powerful pyranopyridine derivatives. *Proc Natl Acad Sci* USA 2016;113:3509–14.

[34] Gervasoni S, Malloci G, Bosin A, Vargiu AV, Zgurskaya HI, Ruggerone P. Recognition of quinolone antibiotics by the multidrug efflux transporter MexB of *Pseudomonas aeruginosa*. Phys Chem Chem Phys 2022;24:16566–75.

[35] Li J, Zhang H, Ning J, Sajid A, Cheng G, Yuan Z, et al. The nature and epidemiology of OqxAB, a multidrug efflux pump. Antimicrob Resist Infect Control 2019;8:1–13.

[36] Zhao J, Chen Z, Chen S, Deng Y, Liu Y, Tian W, et al. Prevalence and dissemination of oqxAB in *Escherichia coli* isolates from animals, farmworkers, and the environment. Antimicrob Agents Chemother 2010;54:4219–24.

[37] Xu S, Chen G, Liu Z, Xu D, Wu Z, Li Z, et al. Site-directed mutagenesis reveals crucial residues in *Escherichia coli* resistance-nodulation-division efflux pump OqxB. Microb Drug Resist 2020;26:550–60.